

Technology Stack

Date	27 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML, CSS, JavaScript, Bootstrap</i>	<i>Creates a responsive and user-friendly interface for students and teachers across different devices.</i>

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Flask is lightweight, easy to develop with, and integrates well with AI/ML models and REST APIs.</i>
3	Database / Data Storage	<i>SQLite (or MySQL if your project uses it)</i>	<i>Stores user accounts, emotion history, learning progress, and recommendations efficiently</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Google Cloud Platform (Vertex AI), Docker</i>	<i>Enables AI model deployment, scalability, and reliable cloud hosting for the application.</i>
5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Supports version control, team collaboration, code backup, and project management.</i>

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Google Gemini API (Vertex AI), OpenCV, Media Pipe</i>	<i>Gemini API</i> provides personalized learning recommendations and explanations based on detected emotions. <i>OpenCV</i> captures and processes facial images, while <i>Media Pipe</i> detects facial landmarks for accurate real-time emotion recognition.